CECS 361 Spring 2026

Stage 3: Second Progress Report

Custom 16-bit RISC SoC for Digital Lock with VGA Status Display

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Progress

Since the first progress report, the project has moved from the planning stage into an early implementation stage. Initially, the work was focused mainly on defining the project concept, identifying the required subsystems, and researching the major topics needed for development, such as CPU architecture, memory-mapped input/output, VGA fundamentals, and verification strategy. Since then, the design has been refined into a more concrete system Architecture. The CPU datapath, memory map, and peripheral interface have now been defined. The CPU datapath, memory map, and peripheral interface have now been defined, and initial modules for the instruction ROM, data RAM, and selected I/O registers have been created. In addition, the lock-conctrol program flow has been drafted, and the VGA approach has been improved by simplifying it so that the CPU writes status codes to a dedicated register instead of directly generating video timing. This change makes the overall system easier to design, test, and integrate.

Another major improvement is that the project is no longer only conceptual. The current prototype now reflects a clearer separation between processor logic, memory, and peripheral control, which is an important step toward full FPGA implementation. Compared to the earlier report, the design now has a stronger structural foundation because the main hardware blocks and their relationships have been organized more clearly. This makes future testing and debugging more manageable and gives the team a better path toward integrating the remaining components into a complete system.

The main challenge at this stage involves turning the architecture into a fully working processor system. One challenge is that the CPU is still incomplete and not yet operating as a fully functional processor. While the structure has been laid out, additional work is still needed on control logic, instruction execution, and reliable interaction between the datapath and peripherals. Another challenge is keeping the design simple enough to implement successfully while still meeting the project goals. This includes deciding how much functionality to include in the CPU, how to handle memory mapped I/O cleanly, and how to integrate all modules without introducing unnecessary complexity

A further challenge is system integration and verification. Even when individual modules compile or appear correct on their own, combining ROM, RAM, registers, input devices, and display outputs into one coordinated design can create timing, control, and debugging difficulties. The VGA portion also remains a challenge because it must eventually work alongside the processor and status system without overcomplicating the design. Overall, the project has made meaningful progress by moving beyond research and into implementation, but the biggest remaining difficulty is completing and validating the processor so that all parts of the digital lock system operate together correctly.